Date: _____ Day: _____

Section 4.6
The Multiplication Rule

What it is?

The conditional Prob rule is used to compute conditional Prob in terms of unconditional Prob. that is

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

If we multiply both sides with P(A) we obtain a formula for computing joint prob in terms of marginal & conditional Prob and is known as Multiplication Rule.

$$P(B \cap A) = P(A) \cdot P(B|A) \text{ or } P(B) \cdot P(A|B)$$

$$P(B \cap A) = P(B) \cdot P(A) \rightarrow \text{If independent}$$

Both rules are variation of each other. On one hand, when joint & Marginal Prob are known or can easily determined directly we use conditional Probability rule to obtain Conditional Prob. while on other hand, when marginal & conditional Probs are known or can be determined directly we use multiplication rule.

Date: _____ Day: _____

E with Replacement:

- Population Size remains same
- Its independent
- For example

  Red Marker = 1

  Blue Marker = 1

  $P(R) = \frac{1}{2}$

  $P(B) = \frac{1}{2}$

Without Replacement:

- Population size Reduce
- It's Dependent.
- For example

  Red Marker = 1

  Blue Marker = 1

  select Red

  $P(R) = \frac{1}{2} = 0.5$

  select Blue

  $P(B) = \frac{1}{1} = 1$

Example Class Examples

Example:1 (From class)

Software build passes unit test with prob 0.90. Given unit test success, integration test pass with prob 0.80. Find prob both stges passs.

Let event U: Unit Test

Let event I: Integration test

Date: _____ Day: _____

Given Data:

$P(U) = 0.90$

$P(I|U) = 0.80$

$P(U \cap I) = ?$

$\therefore P(U \cap I) = P(U) P(I|U)$

$P(U \cap I) = (0.90)(0.80)$

$P(U \cap I) = 0.72$

$P(U \cap I) = 72\%$

Example:2 (From class)

Serve & backup link A server component works with prob 0.98. Given that it works it backup link work with prob 0.95. Find that both works.

Let event S: serve comp works

Let Event B: Backup link works

Given:

$P(S) = 0.98$

$P(B|S) = 0.95$

$P(B \cap S) = ?$

$\therefore P(B \cap C) = P(S) P(B|S)$

$P(B \cap C) = (0.98)(0.95)$

$P(B \cap C) = 0.931$

$P(B \cap C) = 93.1\%$

Date: _____ Day: _____

Example: 3 (From class)

Independent service Failures: Two services fail independently with prob 0.04 & 0.08.

$P(F_1 \cap F_2) = P(F_1)\,P(F_2)$ Let event F_1: Failure 1

$P(F_1 \cap F_2) = (0.04)(0.08)$ Let event F_2: Failure 2

$P(F_1 \cap F_2) = 0.0012$

$P(F_1 \cap F_2) = 0.12\%$

Example: Multiplication Rule For three Event:

$$P(A \cap B \cap C) = P(A)\,P(B|A)\,P(C|A \cap B)$$

- First, calculate prob of A
- Multiply by Prob of B given A
- Multiply by prob of C given both A and B

If independent:

$$P(A \cap B \cap C) = P(A)\,P(B)\,P(C)$$

Example: 4 (From class)

Three stage deployment Process has these prob.

- Validation Succeeds : $P(V) = 0.95$
- Load testing succeeds given validation
  $P(L|V) = 0.90$
- Deployment succeed given both earlier stages
  $P(D|V \cap L) = 0.98$

Date: _____ D

Find Prob that all three Stage succeed.

$\therefore P(V \cap L \cap D) = P(V)\, P(L|V)\, P(D|V \cap L)$

$P(V \cap L \cap D) = (0.95)(0.90)(0.98)$

$P(V \cap L \cap D) = 0.8379$

$P(V \cap L \cap D) = 83.79\ \%$

Example 5:

Independent AI components: Three Indep components work with prob 0.99, 0.97 and 0.98. Find that ALL three work.

Let event C_1: 1st component works

Let event C_2: 2nd Component works

Let event C_3: 3rd component works

$\therefore P(C_1 \cap C_2 \cap C_3) = P(C_1) \cdot P(C_2)\, P(C_3)$

$P(C_1 \cap C_2 \cap C_3) = (0.99)(0.97)(0.98)$

$P(C_1 \cap C_2 \cap C_3) = 0.9411$

$P(C_1 \cap C_2 \cap C_3) = 94.11\ \%$

Example: 6

A Set contains 8 working and 2 defective devices. Two Devices are selected without replacement. Prob both are working.

Date: _____ Day: _____

Let event W_1: First one Device is working

Let event W_2: Second Device is working

Given Data:

Total Devices = 10

Total Working = 8

Total Defective = 2

Without Replacement

↓

The First Device selected is not put back into set

W	D	T
8	2	10

$$P(W_1 \cap W_2) = \frac{8}{10} \times \frac{7}{9} = \frac{56}{90} = 0.6222$$

$$P(W_1 \cap W_2) = 62.22\%$$

Example: 7 (From Class)

Eggs

Bad (3) Good (9)

Select 2 For omled.

- Find both will be good
- Find First is good & 2nd is bad

a:

Let event, G_1: 1st is Good

Let event G_2: 2nd is Good

$$P(G_1 \cap G_2) = \frac{9}{12} \times \frac{8}{11} = \frac{72}{132} = 0.5454$$

$P(G_1 \cap G_2) = 54.54\%$.

b-

Let event G_1: 1st is Good

Let event B_2: 2nd is Bad

$$P(G_1 \cap B_2) = \frac{9}{12} \times \frac{3}{11} = \frac{27}{132} = 0.2045$$

$P(G_1 \cap B_2) = 20.45\%$.

Exercise Example

Exercise: 4.20

Qs page 181

Let event S: Members are senators

Let event D: Members are Democrate

Given Data:

$P(S) = 18.7\% = 0.187$

$P(D|S) = 49\% = 0.49$

Find:-

Prob the selected member is Democratic Senators : $P(D \cap S)$

Date: Day:

$\therefore P(D \cap S) = P(S) \times P(D|S)$

$P(D \cap S) = 0.187 \times 0.49$

$P(D \cap S) = 0.09163$

$P(D \cap S) = 0.092$

$P(D \cap S) = 9.2\%$

Example : 4.21

Qs on Pg 181-182

Without Replacement

Let event F_1 : Fistt std obtain. is Female

Let event M_2 : second std obtain is Male

$\therefore P(F_1 \cap M_2) = P(F_1) \cdot P(M_2 | F_1)$

$P(F_1 \cap M_2) = \left(\frac{23}{40}\right) \cdot \left(\frac{17}{39}\right)$

$P(F \cap M_2) = (0.575)(0.436)$

$P(F \cap M_2) = 0.2507$

$P(F \cap M_2) = 25.1\%$

Example : 4.22

Qs on Pg 183

Selecting Face card : F

" a king : K

" a Heart : H

Date: _____ Day: _____

We know Two event are independent when

$$P(A \cap B) = P(A) \times P(B)$$

Q:

whether event K is independent of F

So must satisfy:

$$P(K \cap F) = P(K) \times P(F)$$

P(K):-

$$P(K) = \frac{4}{52} = 0.077 = \frac{1}{13}$$

P(F)

$$P(F) = \frac{12}{52} =$$

P(K) x P(F):

$$= \frac{4}{52} \times \frac{12}{52}$$

$$= \frac{48}{2704} = \frac{3}{169}$$

P(K∩F):-

$$P(K \cap F) = \frac{4}{52} = \frac{1}{13}$$

$$P(K \cap F) = P(K) \times P(F)$$

$$\frac{1}{13} \neq \frac{3}{169}$$

Date: _____ Day: _____

So, K & F are not independent.

b-

whether K is independent of H

Must satisfy

$$P(K \cap H) = P(K) \times P(H)$$

$$\frac{1}{52} = \frac{4}{52} \times \frac{13}{52}$$

$$\frac{1}{52} = \frac{52}{2704}$$

$$\frac{1}{52} = \frac{1}{52}$$

So K & H are independent of each other.

Example: 4.23

Qs pg 185

Let events

G_1: ball Lands on Green Fist time

B_2: Ball Land on black second time

B_3: ball Land on black Third Time.

Given Data:

Total = 38

Red = 18 Black = 18

Green = 2

Date: Day:

$\therefore P(G_1 \cap B_2 \cap B_3) = P(G_1) \cdot P(B_2) \cdot P(B_3)$

$P(G_1 \cap B_2 \cap B_3) = \left(\frac{2}{38}\right)\left(\frac{18}{38}\right)\left(\frac{18}{38}\right)$

$P(G_1 \cap B_2 \cap B_3) = 0.012$

$P(G_1 \cap B_2 \cap B_3) = 1.2\ \%$

Exercise : 4.6

Question: 4.123

Qs on Pg 185

a-

Conditional :-

This rule is used to calculate conditional Prob.

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Multiplication:

use to calculate Joint Prob.

$$P(A \cap B) = P(A) \times P(B|A)$$

b-

Both rules are mathematically coz they are rearrangements of same formula.

Date: Day:

• Conditional Prob rules use division to find conditional Prob.

• General Multiplication Rule use multiplication to find Joint Prob.

C-

Because they are used in Diff situations:-

• When Joint & Marginal Prob are known we use conditional Prob rule.

• When Marginal & conditional are known we use Multiplication rule.

Question: 4.125

Qs on Pg 186

Let W = Selected adult is a Women

Let event D: select adult suffer from Depression.

Given Data:

P(W/D P(D|W) = 44% = 0.44

P(W) = 52% = 0.52

∴ P(W∩D) = P(W) × P(D|W)

P(W∩D) = 0.52 × 0.44

P(W∩D) = 0.2288

Date: Day:

P(W∩D) = 22.88 %

Appro 22.88% of US adult are women who suffer from holiday depression.

Question: 127

Qs on Pg 185 (Without Replacement)

Let event H : First number picked is 3

Let event K : Second number picked exceeds 4

a-

P(H) :- S = {1, 2, 3, 4, 5, 6}

P(H) = 1/6 = 0.1667

P(H) = 16.67%

b-

P(K|H)

P(K|H) = 2/5 = 0.4

P(K|H) = 40%

Date: _____ Day: _____

c-

$P(H \& K)$

$P(H \cap K) = P(H) \times P(K|H)$

$P(H \cap K) = \frac{1}{6} \times \frac{2}{5} = \frac{2}{30} = 0.0667$

$P(H \cap K) = 6.67\%$

Prob that both number picked are.

d-

Less than 3

Let event B: both number picked are Less than 3

$P(B) = \frac{2}{6} \times \frac{1}{5} = \frac{2}{30} = 0.0667$

$P(B) = 6.67\%$

e-

greather than 3

Let event A: both are greather than 3

$P(A) = \frac{3}{6} \times \frac{2}{5} = \frac{6}{30} = 0.2$

$P(A) = 20\%$

Date: _____ Day: _____

Question: 4.129

a-

Let event $\bar{J}_1$: First student is Junior

Let event S_2 : Second is Senior.

$$P(\bar{J}_1 \cap S_2) = P(\bar{J}_1) \times P(S_2 | \bar{J}_1)$$

$$P(\bar{J}_1 \cap S_2) = \left(\frac{12}{40}\right) \times \left(\frac{7}{39}\right) = \frac{84}{1560}$$

$$P(\bar{J}_1 \cap S_2) = \frac{7}{130} = 0.054$$

$$P(\bar{J}_1 \cap S_2) = 5.4\ \%$$

b-

Let even S_1 = 1st is somophore

Let event S_2 = 2nd is somophore

$$P(S_1 \cap S_2) = P(S_1) \times P(S_2 | S_1)$$

$$P(S_1 \cap S_2) = \left(\frac{15}{40}\right) \times \left(\frac{14}{39}\right) = \frac{210}{1560}$$

$$P(S_1 \cap S_2) = 0.1346$$

$$P(S_1 \cap S_2) = 13.46\ \%$$

Date: _____ Day: _____

d-

Prob that one of students obtained is a Freshman and other a ~~somopho~~ sophomore.

Case 1: First is Freshman & Second somophore.

$$P(F_1 \cap SO_2) = P(F_1) \times P(SO_2 | F_1)$$

$$P(F_1 \cap SO_2) = \left(\frac{6}{40}\right) \times \left(\frac{15}{39}\right) = \frac{90}{1560}$$

$$P(F_1 \cap SO_2) = 0.0577$$

$$P(F_1 \cap SO_2) = 5.77\ \%$$

Case 2: First is somophore & second is Freshman.

$$P(SO_1 \cap F_2) = \left(\frac{15}{40}\right) \times \left(\frac{6}{39}\right) = \frac{90}{1560}$$

$$P(SO_1 \cap F_2) = 0.0577$$

$$P(SO_1 \cap F_2) = 5.77\ \%.$$

Date: Day:

Question: 4.134

qs on Pg 186

a-

$P(R_3)$

$$P(R_3) = \frac{40,379}{98,993} = 0.4079$$

$P(R_3) = 40.79\%$

b-

$P(R_3 | G_1)$

$$P(R_3 | G_1) = \frac{P(R_3 \cap G_1)}{P(G_1)} = \frac{25,888}{70,000}$$

$P(R_3 | G_1) = 0.3698$

$P(R_3 | G_1) = 36.98\%$

c-

No, They are not independent.

$$P(R_3 | G_1) = P(R_3)$$

$$0.3698 \neq 0.4079$$

Hence proved.

Date: _____ Day: _____

d

No, they are not Independent.

For medical school Faculty member is the event that person is Female (G_{12}) independent of event that person is an associate professor (R_2)

For checking:

$$P(G_{12} | R_2) = P(G_{12})$$

$$\frac{P(G_{12} \cap R_2)}{P(R_2)} = P(G_{12})$$

$$\frac{5400}{21732} = \frac{28993}{98993}$$

$$0.248 \neq 0.2929$$

Hence proved they are Dependent.

Question: 4.133 (Qs on Pg 186-187)

a

$P(R_1)$

$P(R_1) = 0.527$

$P(R_1) = 52.7\%$

Date: _____ Day: _____

$P(C_2)$	$P(C_1 \& C_2)$
$P(C_2) = 0.187$ $P(C_2) = 18.7\%$	$P(C_1 \cap C_2) =$

b-

~~For Independent:~~

Special Multiplication Rule states two events are independent iff

$$P(A \cap B) = P(A) \times P(B)$$

so For our qs

$$P(P_1 \cap C_2) = P(P_1) \times P(C_2)$$

$$0.092 = 0.527 \times 0.187$$

$$0.092 \neq 0.0985$$

so P_1 and C_2 are Dependent.

Question: 4.135 (Qs pg 187)

Let event

A : First toss is Head

B : Third is Tail

C : Total head is 1 (↓ no of)

Date:-

a-

P(A):

$$P(A) = \frac{4}{8} = 0.5$$

$$P(A) = 50\%$$

P(B):

$$P(B) = \frac{4}{8} = 0.5$$

$$P(B) = 50\%$$

P(C):

$$P(C) = 3/8 = 0.375$$

$$P(C) = 37.5\%$$

b-

P(B|A)

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{2}{4} = 0.5$$

$$P(B|A) = 50\%$$

c-

For Independent

$$P(A|B) = P(A)$$

$$\frac{P(A \cap B)}{P(B)} = 0.5$$

$$\frac{2/4}{0.5} = 0.5$$

equal so both are independent.

Date: _____ Day: _____

d-

$P(C|A)$

$$P(C|A) = \frac{P(C \cap A)}{P(A)} = \frac{1}{4} = 0.25$$

$P(C|A) = 25\%$

e-

For Independent:

$\therefore P(A|C) = P(A)$

$$\frac{P(A \cap C)}{P(C)} = P(A)$$

$$\frac{1}{3} = \frac{4}{8}$$

$0.333 \neq 0.5$ So independent as not equal.

Question: 4.137 (Os Pg 187)

Let event A_1: First Card is Ace

Let event A_2: Second card is Ace

Total Card = 52

Total Aces = 4

a-

With Replacement (Independent)

$P(A_1 \cap A_2) = P(A_1) \times P(A_2)$

$P(A_1 \cap A_2) = (4/52) \times (4/52)$

$P(A_1 \cap A_2) = 1/169 = 0.0059$

$P(A_1 \cap A_2) = 0.59\%$

b-

Not Replaced (Dependent)

$P(A_1 \cap A_2) = P(A_1) \times P(A_2 | A_1)$

$P(A_1 \cap A_2) = (4/52) \times (3/51)$

$P(A_1 \cap A_2) = 12/2652 = 0.0045$

$P(A_1 \cap A_2) = 0.45\ \%$

Question: 4.139 (QS Pg 187)

Assuming items are Independent

Let event A: Critical 1 item reliable

Let event A': Critical 1 item Fail

Given Data:

Total Item = 748

$P(A) = 99.99\ \% = 0.9999$

$P(A') = 0.0001$

a-

None of item Fail

$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_{748})$

↳ ALL item works

$P(A) = (0.9999)^{748}$

$P(A) = 0.9279$

$P(A) = 92.79\ \%$

Date: Day:

b-

Atleast 1 item Fail

$P(\text{Atleast 1 Fail}) = 1 - P(A)$

$P(A)$ ↓ none Fail

$P(\text{Atleast 1 Fail}) = 1 - 0.9279$

$P(\text{Atleast 1 Fail}) = 0.0721$

$P(\text{Atleast 1 Fail}) = 7.21\%$

c-

Interpret Ans in Part (b)

There is 7.21% probability that atleast one of 748 "Critically 1" item will Fail

Question: 4.141 (os : Pg 187 to 188)

Crashes are Independent

Given Data.

Fist Crash: Driver Age = 16-24

Second Crash: Driver Age = 25-34

Third Crash : Driver Age = 34-64

a:

Let even A: First Driver is 16-24

Let event B: Second Drive is 25-34

Let event C: Third Driver is 35-64

Day:

$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

$P(A \cap B \cap C) = (0.255) \times (0.238) \times (0.393)$

$P(A \cap B \cap C) = 0.0239$

$P(A \cap B \cap C) = 2.39\%$

b-

Let A: Driver is 16-24

Let B: Driver is 65 or older.

Total Arrangements

A X A X B

A X B X A

B X A X A

Case 1: 1st Two in age grp (16-24) and 3rd in 65 or oldre.

$P(A \cap A \cap B) = P(A) \times P(A) \times P(B)$

$P(A \cap A \cap B) = 0.255 \times 0.255 \times 0.114$

$P(A \cap A \cap B) = 0.0074$

$P(A \cap A \cap B) = 0.74\%$

Case 2: For ALL three Arrangement

$P = (AAB) + (A \cap B \cap A) + (B \cap A \cap A)$

$P = 0.0074 + 0.0074 + 0.0074$

$P = 0.0222$

$P = 2.22\%$

Date: _____ Day: _____

Question: 4.143 (Qs Pg 188)

Assuming selections are Independent.

a-

Let event A : Person is 80 y/o or older.

$P(A \cap A \cap A) = P(A) \times P(A) \times P(A)$

$P(A \cap A \cap A) = 0.033 \times 0.033 \times 0.033$

$P(A \cap A \cap A) = 0.0000359$

$P(A \cap A \cap A) = 0.00359\ \%$

b-

Let event B : Three people from Asia is 80 yrs or older.

$P(B \cap B \cap B) = 0.008 \times 0.008 \times 0.008$

$P(B \cap B \cap B) = 0.000000512$

$P(B \cap B \cap B) = 0.0000512\ \%$

c-

Let

event A : First is in 65-74

event B : Second is in 75-79

event C : Third is in 80 or older.

$P(A \cap B \cap C) = P(A) \times P(B) \times P(C)$

$P(A \cap B \cap C) = 0.066 \times 0.027 \times 0.042$

$P(A \cap B \cap C) = 0.0000748$

$P(A \cap B \cap C) = 0.00748\ \%$

Day:

d-

I would have solved using sampling with replacement.
No it makes a very little Difference. Population of elderly people in North America and Asia is very large and we select only 3 people. Therefore removing 3 people makes a negligible difference to Probs.

Question: 4.145 (Qs Pg 188)

Let
event M : selected person is male
event F : selected person is Female
event A : Person has activity Limitation.

Given:
$P(A|M) = 13.6\% = 0.136$
$P(A|F) = 14.4\% = 0.144$

For Independence:
$P(A|M) = P(A)$
$P(A|F) = P(A)$

$P(A|M) = P(A|F)$
$0.136 \neq 0.144$

Date: Day:

No gender and activity Limitation are not Statistically Independent

Question: 4.147 (Qs Pg 188)

Qs on Pg

Without Replacement:

Let

event F_1 : 1st Std is Female

event F_2 : 2nd Std is Female

event M_3 : 3rd Std is Male

$$P(F_1 \cap F_2 \cap M_3) = P(F_1) \times P(F_2 | F_1) \times P(M_3 | (F_1 \cap F_2))$$

$$P(F_1 \cap F_2 \cap M_3) = \left(\frac{23}{40}\right)\left(\frac{22}{39}\right)\left(\frac{17}{38}\right)$$

$$P(F_1 \cap F_2 \cap M_3) = \frac{4301}{29,640} = 0.1451$$

$$P(F_1 \cap F_2 \cap M_3) = 14.51\%$$

Question: 4.149 (Qs Pg 188)

a-

If mutually exclusive

then $P(A \cap B) = 0$

Date: Day:

b- (Positive Prob)

If two nonimpossible events are independent explain why their Prob is not 0.

Independent event Follows

$P(A \cap B) = P(A) \times P(B)$

coz both events have +ive prob (nonimpossible) their prob can't be 0. The

$P(A \cap B) \neq 0$

c-

When rolling a Die:

A: getting an even number: 6

B: getting a number greater than 3: 6

~~P(A~~ 1. For independent

$P(A \cap B) = P(A)$

$P(A \cap B) = P(A)$

$P(B)$

$2/6 \neq 3/6$

So A and B are Dependent.

Date: _____ Day: _____

2. Mutually Exclusive:

A and B are not mutually exclusive coz both can occur together.

$P(A \cap B) = 2/6 = 1/3$

As joint prob is not 0 so they are not mutually exclusive.

Question: 4.151 Qs Pg 189

Reference Qs 4.150.

a-

A, B and are Independent:

Individual Prob.

$P(A) = 3/6 = 1/2$

$P(B) = 3/6 = 1/2$

$P(C) = 18/36 = 1/2$

First Pair $P(A \cap B) = P(A) \times P(B)$

$P(A \cap B) = P(A) \times P(B)$

$\frac{3 \times 3}{36} = 1/2 \times 1/2$

$9/36 = 1/4$

$1/4 = 1/4$

Satisfied.

Date: _____ Day: _____

check second pair : $P(A \cap C) = P(A)\,P(C)$

$P(A \cap C) = P(A) \times P(C)$

$\frac{3 \times 3}{36} = \frac{1}{2} \times \frac{1}{2}$

$\frac{1}{4} = \frac{1}{4}$

Pair
Sum when Both Die Even or odd.

satisfied.

Third Pair (B and C)

$P(B \cap C) = P(B) \times P(C)$

$\frac{3 \times 3}{36} = \frac{1}{2} \times \frac{1}{2}$

$\frac{1}{4} = \frac{1}{4}$

satisfied

ALL three Events Together.

$P(A \cap B \cap C) = P(A)\,P(B) \times P(C)$

$\frac{3 \times 3}{36} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$

$\frac{1}{4} \neq \frac{1}{8}$

Not satisfied

As one condition is not satisfied so they Dependent.

Date: _____ Day: _____

b-

Show
$P(D \cap E \cap F) = P(D)\,P(E)\,P(F)$ but
D, E and F are not independent.

Individual Prob:

$$P(D) = \frac{3}{6} = \frac{1}{2}$$

$$P(E) = \frac{3}{6} = \frac{1}{2}$$

$$P(F) = \frac{4}{36} = \frac{1}{9}$$

Joint of ALL Three

$$P(D \cap E \cap F) = \frac{1}{36}$$

Final:

$$P(D \cap E \cap F) = P(D) \times P(E) \times P(F)$$

$$\frac{1}{36} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{9}$$

$$\frac{1}{36} = \frac{1}{36}$$

Show D, E and F are not Independent:
we need only one condition that Fails.
Let check $P(D \cap E) = P(D) \times P(E)$

$$P(D \cap E) = \frac{6}{36} = \frac{1}{6}$$

↓
both event happen when red Die must be 3
black can be any $\{1,2,3,4,5,6\}$.

Date: _____ Day: _____

So

$P(D \cap E) = P(D) \times P(E)$

$\frac{1}{6} = \frac{1}{2} \times \frac{1}{2}$

$\frac{1}{6} \neq \frac{1}{4}$

So Dependent coz pairwise condition for D and E FAILS even though three event condition satisfied.